
4. Computational visual analysis in political communication

Yilang Peng and Yingdan Lu

Scholars have long investigated the role of visuals in political communication, ranging from visual representations of politicians (Grabe & Bucy, 2009; Joo et al., 2014; Peng, 2019) to images used in protests (Casas & Williams, 2019; Sobolev et al., 2020; Steinert-Threlkeld et al., 2022) to photographs of humanitarian crises (Dumitrescu & Bucy, 2021). Many important questions remain to be fully addressed by political communication scholars: How do people communicate politics through images and videos? What common types of visual propaganda do governments create and disseminate? How are politicians visually represented on social media? The proliferation of visual-based social media platforms (e.g., Instagram, TikTok) has resulted in an explosion of online visual content, which has greatly expanded opportunities for social scientists to understand media production and dissemination. While traditional visual analysis methods, such as manual coding and qualitative reading, are effective but time consuming, recent breakthroughs in computer vision and deep learning enable researchers to examine visual content computationally and automatically.

In this chapter, we provide a systematic review of how automated visual analysis can be applied in political communication. In particular, we focus on three types of techniques: first, we provide an introduction of supervised learning on image data and present case studies showing how it can be applied to study political protests and misinformation. Second, we introduce unsupervised image clustering and explain how it can be used to explore content themes in large-scale visual data. Third, we introduce some off-the-shelf libraries and Application Programming Interfaces (APIs) for computational analysis of objects, faces, emotions, and aesthetics. We also discuss how computer vision techniques have been extended to video analysis to shape the video-as-data paradigm. We conclude this chapter by highlighting ethical and methodological concerns that we urge researchers conducting future research in this field to consider.

SUPERVISED LEARNING ON IMAGE DATA

Automated learning on images centers on two fundamental tasks: classification and detection (Joo & Steinert-Threlkeld, 2018). The goal of image classification is to assign a label to the input image. A classic example is identifying whether an image is of a dog or a cat. The goal of detection, which is more complicated than classification, is to identify the type and location of an object in a given image input. For instance, detecting faces in an image input requires the existence of the faces to be identified and their locations mapped. To classify or detect objects in images, researchers can train a model to predict the outcome (usually a probability) from the input by learning from labeled input–output relationships in training data, which is also known as the “ground truth.” This process of learning from labels pre-assigned to train-

ing data is known as supervised learning. Similar to supervised learning on texts, supervised machine learning on images requires preprocessing of the image data into a “bag of features” for model training and prediction (Loussaief & Abdelkrim, 2016). Common machine learning algorithms, such as support vector machine and random forest, are widely used in supervised learning on images.

Breakthroughs in computer vision and machine learning have enabled researchers to employ better, more complex models to understand images. Deep learning with hierarchical structures consisting of multiple layers is widely used in automated image analysis. Unlike traditional machine learning algorithms, which rely on arbitrary human-selected features (e.g., edges) to predict outcomes, deep learning algorithms, such as convolutional neural networks (CNNs), capture features directly from visuals and represents higher layers by building upon lower layers, thereby surpassing traditional machine learning in image classification and other visual-based learning tasks and saving on computational costs (Joo & Steinert-Threlkeld, 2018; Williams et al., 2020).

Although deep learning algorithms require a large amount of annotated data for supervised learning, it is not necessary to train a new model from scratch. Instead, researchers can copy the top layers from an existing model that has been pretrained on a large-scale annotated image dataset (e.g., ImageNet) to their own model and then train other layers based on their own data to predict the outcome. This procedure is known as fine-tuning (Chu et al., 2016). The philosophy behind the fine-tuning procedure is transfer learning; that is, transferring knowledge (features) learned on one problem to solve another similar problem.¹

An increasing number of political science studies are applying deep learning and transfer learning techniques for supervised image classification and detection. Here, we elaborate on how these techniques are used in two important research domains: political protest and misinformation. While digital technologies change the opportunity structure for protest and lower the costs of constructing alliances for mobilization (Dagaev et al., 2013; Garrett, 2006; McAdam, 1999; Tarrow, 1994), they also provide more opportunities for researchers to analyze political protests using new data sources, such as digital traces. Recent studies have focused more on image-based social media posts as important digital trace data for understanding political protests. Zhang and Pan (2019) implemented deep learning frameworks on both image and text data collected from the Chinese social media platform Weibo. They trained a two-stage classifier capable of determining whether a social media post represents an offline collective action. Steinert-Threlkeld et al. (2022) went a step further by training a CNN model based on human annotation to estimate state and protest violence in protest images, thereby paving a new path toward understanding the dynamics of political protests.

Misinformation and disinformation is another promising area in which supervised learning can be paired with computational image data to advance political science research. While existing studies have mostly focused on text-based features of fake news, such as factual elements, language patterns, and emotional indications, this study focuses on visual misinformation. The spread of visual misinformation and disinformation has posed a new challenge for democracy and informed citizenship around the world (Lazer et al., 2018; Vosoughi et al., 2018). Images can convey visual misinformation in three ways: image manipulation, out-of-context images, and real images with false claims (Cao et al., 2020; Garimella & Eckles, 2017). A growing body of research is using supervised learning with deep learning frameworks on images to detect splices (Pomari et al., 2018) and image–text mismatches (Ha et al., 2020). Advanced research in computer science has used multimodal content as input to predict visual misinfor-

mation using more complicated architectures involving both structural and semantic features (Qi et al., 2019; Wang et al., 2018). In a word, supervised image classification has paved the way for more research to detect political misinformation and disinformation in political communication.

IMAGE CLUSTERING: AN UNSUPERVISED APPROACH TO CATEGORIZING IMAGES

Supervised learning allows researchers to train a prediction model that maps data (e.g., images) into predefined labels or categories. However, this means that researchers often need to prepare a collection of labeled data. In contrast, unsupervised learning is aimed at discovering hidden patterns or categories without labeled data. Unsupervised learning is frequently applied in computational social science. For example, topic modeling is widely used in natural language processing to identify “topics” from a large collection of textual documents, such as news stories, scientific abstracts, social media posts, user comments, and participant responses in surveys and experiments (Lu & Pan, 2021; Murashka et al., 2021; Xu et al., 2020). An unsupervised approach is also useful for exploring or identifying image categories from a pool of unlabeled visual data.

- Image clustering is an unsupervised technique that enables researchers to automatically categorize visual data into similar-looking categories. Image clustering has gained popularity in social science research (Hu et al., 2014; Manikonda & De Choudhury, 2017; Muise et al., 2022) and has been used to discover categories from politicians’ Instagram posts (Peng, 2018), visual themes in Chinese protests (Zhang & Peng, 2022), and activities in mobile screenshots (Muise et al., 2022).
- While there are several possible approaches to image clustering, one technique that is easy to implement and that achieves satisfactory performance is to combine transfer learning (in particular, feature extraction with a pretrained model) with clustering techniques (Zhang & Peng, forthcoming). The first step in this approach is to feed raw images into a pretrained model and extract features from one of the top layers. This step efficiently obtains a low-dimensional representation of the original image while retaining the most important information. For example, we can represent a 600×400 RGB image, which has $2,400,000 \times 3$ data points, into a vector with only 4,096 dimensions with a pretrained VGG16 model. Next, standard clustering techniques (e.g., k-means) can be applied on these low-dimensional representations to generate clusters.
- One important factor to consider in transfer learning is that the chosen pretrained model should be trained with a dataset that is similar to the dataset at hand. Many publicly available pretrained models are trained on the ImageNet Large Scale Visual Recognition Challenge dataset, a popular dataset that contains 1.4 million images of 1,000 object categories. These categories cover a broad range of subjects, including animals, food, and everyday objects, which reflect content themes typically found on social media. Another widely used dataset is Places365, which contains 1.8 million images of 365 scene categories (Zhou et al., 2017).

While k-means clustering categorizes images into similar-looking clusters, researchers can take the further step of combining similar clusters into more meaningful “categories” or

“topics.” To do so, researchers, having used the k-means algorithm to obtain clusters from images, then manually inspect all the clusters, interpret and label them, and combine similar clusters into a larger category (Hu et al., 2014; Manikonda & De Choudhury, 2017; Muise et al., 2022).

COMPUTER VISION LIBRARIES AND APIs

Besides modeling using supervised or unsupervised methods, researchers can also use computer vision libraries and APIs to perform standard computer vision tasks, such as object recognition and facial recognition. These services do not require advanced technical knowledge or impose excessive computational constraints. Here, we introduce some common tools for solving four tasks: object recognition, face detection, recognition, and analysis, emotion detection, and computational aesthetics.

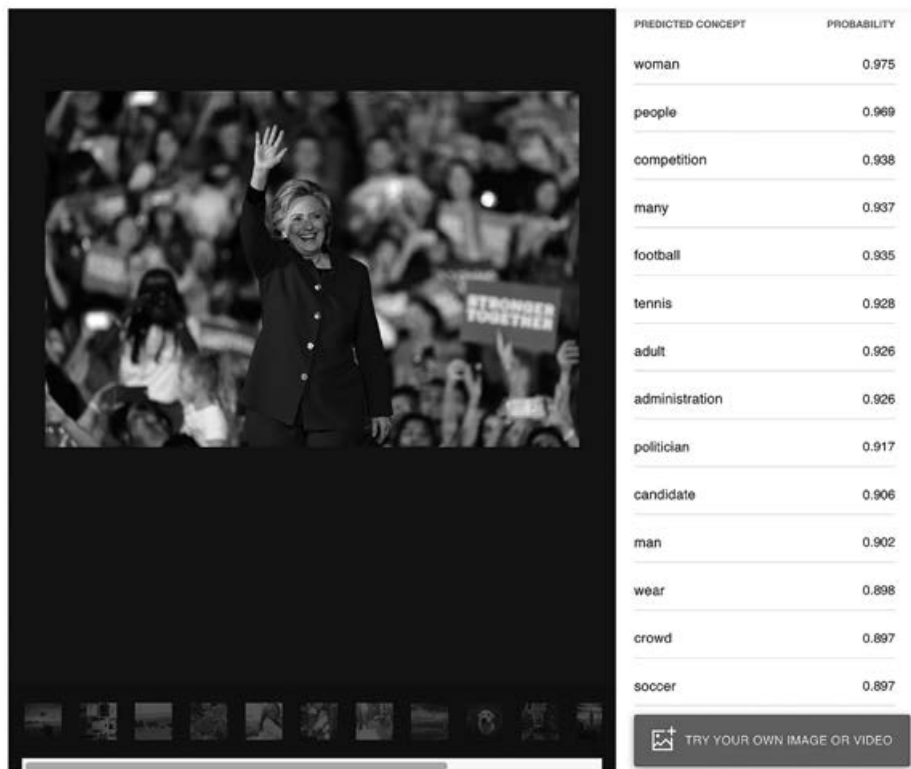
Object Recognition

Object recognition services often provide a list of terms that best describe an image’s content. Figure 4.1 shows the results of an object recognition service—Clarifai—for a picture of Hillary Clinton. At first glance, some tags returned by these services are not entirely accurate: the algorithms did not accurately distinguish between a political gathering and a sporting event. However, the results reveal some valuable information about the pictures, such as whether the politician is surrounded by a crowd. In addition, after extracting descriptive tags for each photo, we can also apply tools of computational linguistic analysis, such as supervised text classification, semantic network analysis, and topic modeling, to make better sense of the data (Araujo et al., 2020; Manikonda & De Choudhury, 2017). This may be a better approach than considering each tag individually. In image classification tasks, CNNs are trained to return a list of tags with the highest probabilities, which are often semantically connected and visually similar. For instance, tags such as *business*, *politician*, and *leader* often appear together, which might be due to the fact that they tend to describe individuals wearing formal suits in professional settings. By grouping semantically similar tags, it is possible to distinguish an overall theme that summarizes a category of tags, even if one particular tag is not particularly accurate.

Face Detection, Recognition, and Analysis

Face detection is a specific type of object recognition that identifies human faces in digital imagery (Wong et al., 2001). Applied in a wide range of contexts, such as surveillance, security, and social media, face detection is one of the most mature computer vision techniques. Researchers today have many commercial face detection services at their disposal, including Face++, Microsoft Azure, and Google Vision. Besides identifying faces, face detection services also return the locations of detected faces in images, usually shown as face rectangles and face landmarks. These mark the positions of important face components, such as the right corner of the left eye (Figure 4.2).

Studies of politicians or political influencers require researchers not only to locate faces in digital imagery but also to recognize the identities of faces. This task is called face recognition



Source: Photo from Gage Skidmore/Flickr. <https://www.flickr.com/photos/gageskidmore/30648686392/in/photolist-NGjwFs-NSC1UF-8xejVb-8xbiLe-Np52mf-4uiqaY-CE1Kff-CVAJZ4-D3TFBa-MULUAL-NSCKJp-hDCf4n-kz57rH-dBACa6-FcyJZ-4UphAs-8QNDRJ-gqhcR3-kz6M6A-4Uk4sK-3qVozN-MpTerx-3k3MrK-3k3Ma8-D1G7iq-Dkxwmk-4vTzFt-BVigj8-BVbh2q-DdvRzR-CE1KRR-BVb2hf-CjbqQK-CH8P43-CQnVbC-3qQPFv-MMbZzG-MMbZdQ-MXgEaT-3qQRbx-DieVvf-DqvKAG-D1G26b-DieN3j-F9hFEx-8xbpv4-8xejXf-GYCzVF-eeoN5G-NPoew1>.

Figure 4.1 *Example of object recognition results*

(Klare et al., 2012). Face recognition web services offer two ways of identifying people in images. The first method is “face verification” or “face search.” Face verification compares two faces and determines whether they belong to the same person, whereas face search compares an input face to other faces in an existing face set and returns the face in the set that best matches the input face. For the latter method, researchers must prepare a face set beforehand that contains all the identities they wish to target. For a new face, the service compares the face to those in the existing face set and returns the identity that best matches the face. This method is often referred to as the “celebrities” mode. Some computer vision services have their own databases of celebrities and public figures. When the user uploads an image to the service, the service detects the face in the image and compares it to the faces in its own database.

Some face detection services also provide a list of facial attributes, such as age, gender, eye openness, hair color, and emotional expression. Some attributes can be calculated based on the



Source: Photo from White House. [https://commons.wikimedia.org/wiki/File:President_of_the_United_States_Joe_Biden_\(2021\).jpg](https://commons.wikimedia.org/wiki/File:President_of_the_United_States_Joe_Biden_(2021).jpg).

Figure 4.2 *Illustration of (a) a face rectangle and (b) face landmarks detected by Face++*

spatial relationships between facial landmarks, such as eye openness. Other attributes, such as age, gender, and emotions, require the algorithms to be trained on a large pre-labeled dataset.

Emotion Detection

Emotion detection is a specific subset of facial analysis that identifies facial expressions in images (Susskind et al., 2007). Most computer vision services use the emotion classification system proposed by Ekman (1992), which categorizes facial expressions into one of six basic emotions—happiness, sadness, disgust, anger, fear, and surprise—or deems them neutral (i.e., absence of emotion). However, some services do not follow this convention. For example, Microsoft Azure has an additional emotion category called “contempt,” and Google Vision does not include the categories “fear” and “disgust.”

Computational Aesthetics

Visual aesthetics shape a variety of psychological and communication outcomes, such as aesthetic appeal, emotion, psychological distances, and persuasiveness (Peng, 2017; Peng & Jemmott, 2018; Sutton et al., 2019). In addition to identifying the content of images, computer vision tools can extract aesthetic features, such as brightness, contrast, composition, color, texture, blur, and complexity (Bakhshi & Gilbert, 2015; Liu et al., 2016; Peng & Jemmott, 2018). These features are often unrelated to the actual content of images, but they can still meaningfully impact our perceptions of images. For example, Peng and Jemmott (2018) demonstrated that computationally calculated features, such as arousing or relaxing colors, feature complexity, and color variety, could successfully predict the aesthetic appeal and likability of food images. Peng (2022) developed *Athec*, a Python library that calculates a wide range of aesthetic attributes that apply to images.

Accuracy and Bias

It is important to keep in mind the potential inaccuracies of different computer vision applications. Face detection is a relatively mature technique with satisfactory accuracy. For example, Abdullah et al. (2015) claimed that the algorithms used in their study achieved an accuracy of 100% in discovering faces in their test dataset. In a study on Instagram photos (Bakhshi et al., 2014), the percentage of agreement between results from a face detection service (Face++) and crowdsourced workers' judgments was 97% for whether an image has a face or not, 96% for having a female or a male face, and 93–99% for matching a face to an age group.

Regarding emotion detection, the currently available computer vision tools can satisfactorily detect the presence of positive emotions, such as happiness, but are not accurate in distinguishing between other types of facial expressions. For example, the computational algorithm used by Abdullah et al. (2015) detected smiles in images with a high accuracy rate (area under the ROC curve = 0.9). In another study of visual portrayals of politicians, computational detection of happiness, anger, and emotional neutrality corresponded relatively well to crowdsourced workers' evaluations, but the correspondence was not so good for other emotions, such as fear (Peng, 2018).

There may also be potential biases in face detection and facial recognition algorithms in relation to different demographic groups (Buolamwini & Gebru, 2018; Klare et al., 2012; Phillips et al., 2011; O'Toole et al., 2012). The accuracy of face recognition algorithms largely depends on the training dataset. If the training dataset predominantly comprises faces from a certain demographic group, the algorithms trained on this specific dataset might not be particularly accurate for faces from other demographic groups (Klare et al., 2012). A study comparing face recognition algorithms from different countries showed that algorithms developed in East Asian countries were better at discerning Asian faces, whereas algorithms developed in Western countries were better at recognizing Caucasian faces (Phillips et al., 2011). An analysis of six facial recognition algorithms showed that the algorithms had lower accuracy for females, Blacks, and individuals in the 18–30 age group (Klare et al., 2012).

FROM IMAGE-AS-DATA TO VIDEO-AS-DATA

Along with the surge in online image data, the development of digital video technologies and the proliferation of video-sharing platforms, such as YouTube and TikTok, have led social scientists to turn their attention to numerous video data sources. Unlike static images, videos afford more modalities (visual, audio, and text) and can thus provide direct portrayals of reality and stimulate human cognitive systems more effectively (Marmolin, 1992; Sundar, 2000). This means that videos can elicit more comprehensive emotional responses than static images (Iyer & Oldmeadow, 2006). Furthermore, the larger number of realism cues embedded in videos may lead to a higher level of trust (Hoogeveen, 1997; Sundar, 2008). The effects of videos have been widely studied in relation to different topics of political communication, such as information learning from television news (Edwardson et al., 1981; Graber, 1990; Reese, 1984), the persuasive effects of video-based political advertisements (Daignault et al., 2013), the role of video-based propaganda in shaping public opinion (Pan et al., 2022), and the effects of videos in correcting false beliefs (Young et al., 2018). However, most political communication studies of videos have applied either qualitative or experimental designs,

which cannot afford quantitative analyses of video data at scale and prove costly for obtaining useful data.

Recent research on political communication has focused on automated analysis of videos to overcome the drawbacks of human coding. We summarize this research into two streams: single-modality analysis and multimodal analysis. The first research stream focuses on a single modality, either the visual, auditory, or text modality. As videos can be broken down into consecutive visual frames, studies of videos often apply frame-level visual analysis of aesthetic or facial features. By detecting and analyzing faces from sampled frames of television videos, Jürgens et al. (2022) found a significant underrepresentation of women and elderly people on German TV. Kang et al. (2020) extracted body movements from frames of debate videos and found a correlation between body gestures and facial emotions while candidates were speaking, and Dietrich (2021) used frame differencing and motion detection to measure the intensity of motions of the video frames and found it to be predictive of party votes. Recent breakthroughs in computer vision have brought more state-of-the-art visual-based models and tools for handling various video analysis tasks, such as video-by-video comparison using ViSiL (Kordopatis-Zilos et al., 2019) or SSAN (Jiang et al., 2021) and pretrained models for visual-based video classification, such as the InceptionV3 model (Szegedy et al., 2016). It is likely that these models will be exploited further by social scientists to solve research questions concerning political communication.

In addition to the analysis of visual features, one strand of recent political science research has applied an audio-as-data framework to extract auditory features, such as pitch and speaker, from videos of political speeches (Dietrich & Juelich, 2018; Dietrich et al., 2019; Neumann, working paper) to investigate the manifestations of these features and their effects on political activities. Another common practice for understanding videos is video transcript analysis, which transforms a video-understanding task into an NLP task. Cochrane et al. (2022) found that transcripts of political speeches can capture the same sentiments as videos. They also found that the word-embedding approach surpassed both the dictionary method and supervised machine learning as an automated method for understanding the sentiment of Canadian parliamentary speeches.

Although analyses that focus on a single modality can determine certain visual, auditory, or textual features of videos, such single-modality frameworks provide a relatively narrow understanding of videos. Therefore, another strand of research has explored how multiple modalities can be analyzed simultaneously to acquire a more comprehensive understanding of videos. Lu and Pan (2022) proposed a multimodal analytical framework to analyze more than 50,000 propaganda videos on Chinese TikTok. They combined computer-vision-based analysis of visual features in video frames, textual descriptions of videos, and account-level analysis of metadata affiliated with each video. Rittmann et al. (2020) used both audio and visual inputs to train a CNN to predict the emphasis in legislative speeches. Advanced computer vision frameworks, such as AttentionVLAD (Li et al., 2021), go a step further by combining visual, auditory, and textual features to predict the identity of people in videos. This kind of synthesized multimodal framework can be fine-tuned and applied to other classification or detection tasks related to political communication, such as distinguishing propaganda videos from non-propaganda videos and detecting political leaders in videos.

DISCUSSION

In this chapter, we introduce automated visual analysis via supervised learning, unsupervised learning, and state-of-the-art computer vision tools and highlight how the image-as-data paradigm has been extended to include video analysis. While the surge in image and video data has created more opportunities for political communication researchers, we highlight the potential ethical and methodological concerns that researchers in this field should consider when using computational visual analysis tools. Williams et al. (2020) identified three ethical concerns that should be considered: privacy protection, potential harm, and social algorithmic bias. The collection, storage, and analysis of visual data all entail privacy risks. For instance, the collection of visual data without consent may violate the privacy terms of some social media platforms. Also, visuals like social media traces may include identifying information that are unconsciously revealed, which may be more problematic when the data are stored and analyzed via cloud services or online APIs based on cloud computation.

Visual analysis can also inadvertently cause harm. For instance, research algorithms or models developed to identify protestors may be used by authoritarian regimes to advance their surveillance tools and repress protests. When releasing open datasets, researchers must also consider whether the images or videos in the dataset can cause harm to those who may access them. In addition, when training models are used to classify potentially harmful visuals, such as deepfakes or pornography, through supervised machine learning, researchers must protect the human annotators from being “polluted” or hurt by the visuals.

Finally, as algorithms are trained using datasets, it is possible that any inherent biases within the training datasets could be carried over to the algorithms. Shankar et al. (2017) found an “amerocentric and eurocentric representation bias” in two mainstream open-source image datasets and showed how this geographical bias can result in misclassification problems (p. 1). The Gender Shades project by Buolamwini and Gebru (2018) shows that mainstream gender classification algorithms perform best on males and subjects with lighter skin tones and worst when classifying females with darker skin. Therefore, before using any open-source dataset or pretrained model, researchers should assess whether the given dataset or model fits their research questions and rule out potential biases.

Besides these ethical concerns, we also urge researchers in the field of computational visual analysis to pay more attention to method validation. This is especially important in relation to the use of state-of-the-art computer vision libraries or APIs, when the ease of acquiring outputs may be prioritized over the model performance. As we highlighted in our discussion of computer vision libraries and APIs, tools for detecting facial expressions need to be improved to target certain emotions. Therefore, we urge researchers to perform human validation after applying off-the-shelf tools to obtain a direct sense of how well the algorithm works. We also encourage researchers to use different algorithms for the same task and compare the results to find the most suitable tool with the best performance.

NOTE

1. For more details, see https://keras.io/guides/transfer_learning.

REFERENCES

- Abdullah, S., Murnane, E. L., Costa, J. M., & Choudhury, T. (2015, February). Collective smile: Measuring societal happiness from geolocated images. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing* (pp. 361–374).
- Araujo, T., Lock, I., & van de Velde, B. (2020). Automated visual content analysis (AVCA) in communication research: A protocol for large scale image classification with pre-trained computer vision models. *Communication Methods and Measures*, 14(4), 239–265.
- Bakhshi, S., & Gilbert, E. (2015). Red, purple and pink: The colors of diffusion on Pinterest. *PloS one*, 10(2), e0117148.
- Bakhshi, S., Shamma, D. A., & Gilbert, E. (2014, April). Faces engage us: Photos with faces attract more likes and comments on Instagram. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 965–974).
- Buolamwini, J., & Gebru, T. (2018, January). Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Conference on Fairness, Accountability and Transparency* (pp. 77–91). PMLR.
- Cao, J., Qi, P., Sheng, Q., Yang, T., Guo, J., & Li, J. (2020). Exploring the role of visual content in fake news detection. *Disinformation, Misinformation, and Fake News in Social Media*, 141–161. <https://doi.org/10.48550/arXiv.2003.05096>
- Casas, A., & Williams, N. W. (2019). Images that matter: Online protests and the mobilizing role of pictures. *Political Research Quarterly*, 72(2), 360–375.
- Chu, B., Madhavan, V., Beijbom, O., Hoffman, J., & Darrell, T. (2016, October). Best practices for fine-tuning visual classifiers to new domains. In *European Conference on Computer Vision* (pp. 435–442). Springer.
- Cochrane, C., Rheault, L., Godbout, J. F., Whyte, T., Wong, M. W. C., & Borwein, S. (2022). The automatic analysis of emotion in political speech based on transcripts. *Political Communication*, 39(1), 98–121.
- Dagaev, D., Lamberova, N., Sobolev, A., & Sonin, K. (2013). Technological foundations of political instability. CEPR Discussion Paper No. DP9787. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2444785.
- Daignault, P., Soroka, S., & Giasson, T. (2013). The perception of political advertising during an election campaign: A measure of cognitive and emotional effects. *Canadian Journal of Communication*, 38(2), 167–186.
- Dietrich, B. J. (2021). Using motion detection to measure social polarization in the US House of Representatives. *Political Analysis*, 29(2), 250–259.
- Dietrich, B. J., & Juelich, C. L. (2018). When presidential candidates voice party issues, does Twitter listen? *Journal of Elections, Public Opinion and Parties*, 28(2), 208–224.
- Dietrich, B. J., Hayes, M., & O'Brien, D. Z. (2019). Pitch perfect: Vocal pitch and the emotional intensity of congressional speech. *American Political Science Review*, 113(4), 941–962.
- Dumitrescu, D., & Bucy, E. P. (2021). Visual portrayals of people in need: The impact of refugee depictions, compassion, and support for humanitarian aid. *International Journal of Public Opinion Research*, 33(4), 818–839.
- Edwardson, M., Grooms, D., & Proudlove, S. (1981). Television news information gain from interesting video vs. talking heads. *Journal of Broadcasting & Electronic Media*, 25(1), 15–24.
- Ekman, P. (1992). An argument for basic emotions. *Cognition and Emotion*, 6, 169–200.
- Garimella, K., & Eckles, D. (2017). Image based misinformation on WhatsApp. In *Proceedings of the Thirteenth International AAAI Conference on Web and Social Media (ICWSM 2019)*.
- Garrett, R. K. (2006). Protest in an information society: A review of literature on social movements and new ICTs. *Information, Communication and Society*, 9(2), 202–224.
- Grabe, M. E., & Bucy, E. P. (2009). *Image Bite Politics: News and the Visual Framing of Elections*. Oxford University Press.
- Graber, D. A. (1990). Seeing is remembering: How visuals contribute to learning from television news. *Journal of Communication*, 40(3), 134–155.
- Ha, Y., Park, K., Kim, S. J., Joo, J., & Cha, M. (2020). Automatically detecting image–text mismatch on Instagram with Deep Learning. *Journal of Advertising*, 50(1), 52–62.

- Hoogeveen, M. (1997). Toward a theory of the effectiveness of multimedia systems. *International Journal of Human-Computer Interaction*, 9(2), 151–168.
- Hu, Y., Manikonda, L., & Kambhampati, S. (2014, May). What we Instagram: A first analysis of Instagram photo content and user types. In *Eighth International AAAI Conference on Weblogs and Social Media*.
- Iyer, A., & Oldmeadow, J. (2006). Picture this: Emotional and political responses to photographs of the Kenneth Bigley kidnapping. *European Journal of Social Psychology*, 36(5), 635–647.
- Jiang, C., Huang, K., He, S., Yang, X., Zhang, W., Zhang, X., ... & Chu, W. (2021, October). Learning segment similarity and alignment in large-scale content based video retrieval. In *Proceedings of the 29th ACM International Conference on Multimedia* (pp. 1618–1626).
- Joo, J., & Steinert-Threlkeld, Z. C. (2018). Image as data: Automated visual content analysis for political science. arXiv preprint arXiv:1810.01544.
- Joo, J., Li, W., Steen, F. F., & Zhu, S. C. (2014). Visual persuasion: Inferring communicative intents of images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 216–223).
- Jürgens, P., Meltzer, C., & Scharrow, M. (2022). Age and gender representation on German TV: A longitudinal computational analysis. *Computational Communication Research*, 4(1), 173–207.
- Kang, Z., Indudhara, C., Mahorker, K., Bucy, E. P., & Joo, J. (2020, August). Understanding political communication styles in televised debates via body movements. In *European Conference on Computer Vision* (pp. 788–793). Springer.
- Klare, B. F., Burge, M. J., Klontz, J. C., Bruegge, R. W. V., & Jain, A. K. (2012). Face recognition performance: Role of demographic information. *IEEE Transactions on Information Forensics and Security*, 7(6), 1789–1801.
- Kordopatis-Zilos, G., Papadopoulos, S., Patras, I., & Kompatsiaris, I. (2019). Visil: Fine-grained spatio-temporal video similarity learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 6351–6360).
- Lazer, D. M., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., & Zittrain, J. L. (2018). The science of fake news. *Science*, 359(6380), 1094–1096.
- Li, F., Wang, W., Liu, Z., Wang, H., Yan, C., & Wu, B. (2021, June). Frame aggregation and multi-modal fusion framework for video-based person recognition. In *International Conference on Multimedia Modeling* (pp. 75–86). Springer.
- Liu, L., Preotiuc-Pietro, D., Samani, Z. R., Moghaddam, M. E., & Ungar, L. (2016, March). Analyzing personality through social media profile picture choice. In *Tenth International AAAI Conference on Web and Social Media*.
- Loussaief, S., & Abdelkrim, A. (2016, December). Machine learning framework for image classification. In *2016 7th International Conference on Sciences of Electronics, Technologies of Information and Telecommunications (SETIT)* (pp. 58–61). IEEE.
- Lu, Y., & Pan, J. (2021). Capturing clicks: How the Chinese government uses clickbait to compete for visibility. *Political Communication*, 38(1–2), 23–54.
- Lu, Y., & Pan J. (2022). The pervasive presence of Chinese government content on Douyin trending videos. *Computational Communication Research*, 4(1), 68–98.
- Manikonda, L., & De Choudhury, M. (2017, May). Modeling and understanding visual attributes of mental health disclosures in social media. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 170–181).
- Marmolin, H. (1992). Multimedia from the perspectives of psychology. *Multimedia*, 39–52. Springer.
- McAdam, D. (1999). *Political Process and the Development of Black Insurgency, 1930–1970*. University of Chicago Press (2nd edition).
- Muise, D., Lu, Y., Pan, J., & Reeves, B. (2022). Selectively localized: Temporal and visual structure of smartphone screen activity across Media Environments. *Mobile Media & Communication*, 2050157922108033. <https://doi.org/10.1177/20501579221080333>
- Murashka, V., Liu, J., & Peng, Y. (2021). Fispiration on Instagram: identifying topic clusters in user comments to posts with objectification features. *Health Communication*, 36(12), 1537–1548.
- Neumann, M. (working paper). Hooked with phonetics: The strategic use of style-shifting in political rhetoric. https://markusneumann.github.io/files/Neumann_APSA.pdf.

- O'Toole, A. J., Phillips, P. J., An, X., & Dunlop, J. (2012). Demographic effects on estimates of automatic face recognition performance. *Image and Vision Computing*, 30(3), 169–176.
- Pan, J., Shao, Z., & Xu, Y. (2022). How government-controlled media shifts policy attitudes through framing. *Political Science Research and Methods*, 10(2), 1–16.
- Peng, Y. (2017). Time travel with one click: Effects of digital filters on perceptions of photographs. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 6000–6011).
- Peng, Y. (2018). Same candidates, different faces: Uncovering media bias in visual portrayals of presidential candidates with computer vision. *Journal of Communication*, 68(5), 920–941.
- Peng, Y. (2019). *Identifying media bias with computer vision*. [Doctoral dissertation, University of Pennsylvania]. Publicly Accessible Penn Dissertations. <https://repository.upenn.edu/edissertations/3508>.
- Peng, Y. (2022). Athec: A python library for computational aesthetic analysis of visual media in social science research. *Computational Communication Research*, 4(1), 323–349.
- Peng, Y., & Jemcott III, J. B. (2018). Feast for the eyes: Effects of food perceptions and computer vision features on food photo popularity. *International Journal of Communication*, 12.
- Phillips, P. J., Jiang, F., Narvekar, A., Ayyad, J., & O'Toole, A. J. (2011). An other-race effect for face recognition algorithms. *ACM Transactions on Applied Perception (TAP)*, 8(2), 1–11.
- Pomari, T., Ruppert, G., Rezende, E., Rocha, A., & Carvalho, T. (2018, October). Image splicing detection through illumination inconsistencies and deep learning. In *2018 25th IEEE International Conference on Image Processing (ICIP)* (pp. 3788–3792). IEEE.
- Qi, P., Cao, J., Yang, T., Guo, J., & Li, J. (2019, November). Exploiting multi-domain visual information for fake news detection. In *2019 IEEE International Conference on Data Mining (ICDM)* (pp. 518–527). IEEE.
- Reese, S. D. (1984). Visual-verbal redundancy effects on television news learning. *Journal of Broadcasting & Electronic Media*, 28(1), 79–87.
- Rittmann, O., Ringwald, T., & Nyhuis, D. (2020). Pay attention to this! Explaining emphasis in legislative speech using automated video analysis in the US House of Representatives. <https://cboussalis.github.io/publication/2020-12-01-rittmann/2020-12-01-rittmann.pdf>.
- Shankar, S., Halpern, Y., Breck, E., Atwood, J., Wilson, J., & Sculley, D. (2017). No classification without representation: Assessing geodiversity issues in open data sets for the developing world. arXiv preprint arXiv:1711.08536.
- Sobolev, A., Chen, M. K., Joo, J., & Steinert-Threlkeld, Z. C. (2020). News and geolocated social media accurately measure protest size variation. *American Political Science Review*, 114(4), 1343–1351.
- Steinert-Threlkeld, Z. C., Chan, A. M., & Joo, J. (2022). How state and protester violence affect protest dynamics. *The Journal of Politics*, 84(2). <https://doi.org/10.1086/715600>
- Sundar, S. S. (2000). Multimedia effects on processing and perception of online news: A study of picture, audio, and video downloads. *Journalism & Mass Communication Quarterly*, 77(3), 480–499.
- Sundar, S. S. (2008). *The Main Model: A Heuristic Approach to Understanding Technology Effects on Credibility*. MacArthur Foundation Digital Media; Learning Initiative.
- Susskind, J. M., Littlewort, G., Bartlett, M. S., Movellan, J., & Anderson, A. K. (2007). Human and computer recognition of facial expressions of emotion. *Neuropsychologia*, 45(1), 152–162.
- Sutton, J. A., Yang, S., & Cappella, J. N. (2019). Perceived effectiveness of objective features of pictorial warning messages. *Tobacco Control*, 28(e1), e24–e30.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2818–2826).
- Tarrow, S. (1994). *Power in Movement: Social Movements, Collective Action and Mass Politics*. Cambridge University Press.
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146–1151.
- Wang, Y., Ma, F., Jin, Z., Yuan, Y., Xun, G., Jha, K., ... & Gao, J. (2018, July). Eann: Event adversarial neural networks for multi-modal fake news detection. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 849–857).
- Williams, N. W., Casas, A., & Wilkerson, J. D. (2020). *Images as Data for Social Science Research: An Introduction to Convolutional Neural Nets for Image Classification*. Cambridge University Press.

- Wong, K. W., Lam, K. M., & Siu, W. C. (2001). An efficient algorithm for human face detection and facial feature extraction under different conditions. *Pattern Recognition*, 34(10), 1993–2004.
- Xu, K., Liu, F., Mou, Y., Wu, Y., Zeng, J., & Schäfer, M. S. (2020). Using machine learning to learn machines: A cross-cultural study of users' responses to machine-generated artworks. *Journal of Broadcasting & Electronic Media*, 64(4), 566–591.
- Young, D. G., Jamieson, K. H., Poulsen, S., & Goldring, A. (2018). Fact-checking effectiveness as a function of format and tone: Evaluating FactCheck.org and FlackCheck.org. *Journalism & Mass Communication Quarterly*, 95(1), 49–75.
- Zhang, H., & Pan, J. (2019). Casm: A deep-learning approach for identifying collective action events with text and image data from social media. *Sociological Methodology*, 49(1), 1–57.
- Zhang, H., & Peng, Y. (2022). Image clustering: An unsupervised approach to categorize visual data in social science research. *Sociological Methods & Research*. doi: 10.1177/00491241221082603
- Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., & Torralba, A. (2017). Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6), 1452–1464.